



Vinamra Gupta

Roll No.: B24CM1068
Bachelor of Technology
Artificial Intelligence and Data Science
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CodeForces | LeetCode
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EDUCATION

Degree/Certificate	Institute/School	CGPA/%	Year
B.Tech in AIDS	Indian Institute of Technology, Jodhpur	7.92/10.0	2024 - Present
Senior Secondary	Little Flowers Public School, Delhi	88.6%	Apr 2022 - May 2024
Secondary	Little Flowers Public School, Delhi	95.2%	Apr 2020 - Mar 2022

EXPERIENCE

- Undergraduate Researcher** [GitHub](#)
Dep. of CSE, IITJ (Supervisor: Dr. Angshuman Paul) May 2025 - Present
 - Conceived a **wavelet-domain MRI denoising framework** using **Rician noise simulation**, expediting restoration of **high-NSA MRI images** from fast, **low-NSA acquisitions**, reducing simulated scan times by up to **75%**.
 - Evaluated **6 deep learning paradigms** including **CNNs, AEs, VAEs, Diffusion Models, and Transformer variants** (DWT/SWT-based); yielded **32.53 PSNR, 0.82 SSIM, 0.05 LPIPS, and 17.12 FID** on **1 to 8 NSA recovery** with **60% recovery** on downstream segmentation tasks, outperforming **5 current SOTA models**.
- Project Member (Inter IIT Tech Meet 14.0 - 7th Place)** [GitHub](#)
Ebullient Securities Challenge (Quant High Preparation Problem Statement) Oct 2025 - Dec 2025
 - Formulated a **quantitative trading strategy** on mid-frequency, **anonymized multi-feature time-series data** under market constraints, processing **over 200 masked features** spanning price, volatility, volume and bands.
 - Streamlined multiple **adaptive indicators and regime-switching models**, including **volatility-based KAMA** and **Kalman filters**; generated **21% and 32% annualized returns** delivering **Calmar ratios of 5.33 and 14.40** with **max drawdowns restricted to 3.83% and 2.18%** on blind out-of-sample data.

PROJECTS

- Food Image Classification** | [GitHub](#) | scikit-learn, PCA, Python Feb 2026 - Apr 2026
 - Accelerated a **4-phase classical ML pipeline** for **20-class food image classification**, integrating **4 handcrafted features** (HOG, LBP, GLCM, Histogram) compressed by **80% via PCA**, alongside a custom **Bayesian KDE** and **7 GridSearchCV-tuned classifiers** evaluated via **stratified 5-fold cross-validation**.
- Trading Strategy** | [GitHub](#) | [Idea](#) | [Performance](#) | Python, Finance, Backtesting Oct 2025 - Dec 2025
 - Architected a **Python-driven backtesting pipeline** evaluating **>500,000 data points**, integrating **volatility-adjusted position-sizing and automated risk controls** to navigate masked price-volume behaviors and **cap maximum drawdown at 2-3%** even during severe market downturns.
- CashFlow Minimizer-Visualizer** | [GitHub](#) | [Deployed Link](#) | C++, Graph Algorithms Sep 2025 - Nov 2025
 - Produced a **transaction optimization platform**, trims redundant cash flows by **90%** using **graph algorithms**, featuring a **real-time Web Worker-driven dashboard** to benchmark **5 algorithms** (**Greedy, Min Cash Flow, Heap, Priority Queue, and Sorting-based**) on execution latency and efficiency with interactive visualizations.

PUBLICATIONS

- PraFormer: Hierarchical Wavelet-Domain Transformer for Single-Shot MRI Denoising**
Vinamra Gupta, Parneeka Makde, Dr. Nivedita, Dr. Subinay Saha, Dr. Angshuman Paul
Submitted for review at **IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) 2027**

ACHIEVEMENTS

- Specialist at Codeforces** (Max Rating: **1544**).
- 7th rank** among **23 IITs** in **Inter IIT Tech Meet 14.0** (Ebullient Securities (Quant PS)).
- Solved **600+ DSA** problems across **Codeforces** and **LeetCode**.
- JEE Advanced 2024**, Secured **AIR 3048**.

TECHNICAL SKILLS

- Languages:** C, C++, Python, JavaScript, React, Vite, HTML, CSS, Data Structures, Graph Algorithms
- Machine Learning & AI:** PyTorch, Pandas, Numpy, LLM fine-tuning (LoRA, QLoRA), TensorFlow, scikit-learn
- Tools:** Git, GitHub, VS Code, Ubuntu, Jupyter Notebook, Colab, Hugging Face, Linux

RELEVANT COURSEWORK

- Data Structures & Algorithms
- Computer Architecture
- Probability & Statistics
- Operating Systems
- Computer Networks
- Machine Learning
- Linear Algebra
- Intro to Computer Science

LEADERSHIP & CAMPUS INVOLVEMENT

- Project Mentor, QuantxRAID Labs**, Mentoring a project on **Financial Machine Learning**.
- Core Member, Quant Labs**, Co-organized campus-wide **hackathons**, conducted **discussions and sessions** for first-year students.
- Assistant Head, Prometeo'26**, directed the planning and execution of **technical events** on the **Unstop platform**.